Region-Grounded and Uncertainty-Aware Visual Question Answering on Medical Images Using General and Medical Vision-Language Models

Abisha K M, Anusiya S, Bhavya Sree P R, and Dirisala Kavya
Department of Artificial Intelligence and Data Science, RMK Engineering College
Kavaraipettai, Tamil Nadu, India
abishakm1507@gmail.com, anusiya2705@gmail.com,
bhavyasree.050305@gmail.com, kavyareddydirisal@gmail.com

Abstract*–Vision-language models (VLMs) can generate fluent radiology-style language to explain their answers, but a fluent answer is not necessarily a valid one. VLMs make claims that the image does not support, often give no reason for them, and tend to be overconfident without ever expressing doubt. We built a web-based application for medical image question answering in which every claim must be backed by an image region, and the system is free to say "I don't know." Segmentation and detection models (MedSAM, YOLO or DETR) first propose numbered candidate regions. The VLM then writes its answer in JSON, attaching a region ID to each finding, and any finding without a valid region is discarded. A confidence gate that combines sampling agreement, predictive entropy and region score decides whether the answer is returned or the case is reported as uncertain. The pipeline wraps Qwen2.5-VL-3B and MedGemma-4B behind the same interface, so they can be compared under identical prompts on SLAKE and VQA-RAD, measuring unsupported-claim rate, pointing accuracy, calibration and robustness. We present the design and implementation of the system and the evaluation protocol. Measured results are not yet reported.

***Index Terms*–medical visual question answering, vision-language models, visual grounding, uncertainty estimation, selective prediction, hallucination**

I. INTRODUCTION

A plain-language interface to medical imaging is highly appealing. A medical student could check a reading against a chest film, a clinician could triage a backlog of scans, and a researcher could mine a dataset, all without writing code. The arrival of vision-language models (VLMs) has made this look feasible, with general-purpose families such as Qwen2.5-VL [4] and medically adapted models such as LLaVA-Med [3] and MedGemma [5].

That said, a chief limitation is that a VLM will answer a question whether or not it has seen the evidence. Hallucination is a well-known quirk of generative models [10], and a benchmark for medical VQA shows that it carries over to clinical images [11]. Three problems follow. A model may describe an abnormality that is not in the image. Its answers are plain text with no spatial link to the image, so the reader has to scour the whole image to corroborate them. And the model will almost never say it is not certain. It is also unclear whether a model pretrained on medical data is more trustworthy than a general-purpose one, or merely more fluent about medicine.

We begin with a simple rule: the system can only claim what it can point to. We turn the rule into three steps. First, we localize the image, forming a set of numbered candidate regions before the language model says anything. Next, the model must ground its findings by citing one of the region IDs, and findings that are not grounded are removed. Finally, the system may abstain, returning "uncertain", when its confidence signals are weak. Consider the question "Is there a pleural effusion?" An answer is kept only if it is grounded by citing a region such as R2 in the right lower zone. A claimed cardiomegaly with no supporting region would be dropped.

This paper presents the design of that system as a proof-of-concept web application and describes our plans for its evaluation. Our main contributions are the following: (i) a machine-checkable pipeline of localization, grounding and abstention; (ii) a common interface for comparing a general and a medical VLM under identical prompts; and (iii) evaluation criteria that go beyond accuracy, including unsupported-claim rate, pointing accuracy, calibration and perturbation tests, plus blinded expert review. Section II reviews related work, Section III covers the method, Section IV the implementation, and Section V our evaluation plan. Section VI discusses limitations and the current status of the work.

II. RELATED WORK

A. Medical Visual Question Answering

Early work in this field used small, manually annotated datasets. VQA-RAD contains questions about radiology images written by clinicians [1]. SLAKE adds pixel-level masks, bounding boxes and a knowledge graph to its question–answer pairs [2]. The region annotations are what interest us here. They let an answer be verified against the actual location of a structure instead of against a label string, which sets SLAKE apart from VQA-RAD. More recent work prompts or fine-tunes large VLMs. LLaVA-Med adapts a general assistant to biomedical figures and captions [3], while MedGemma is an open multimodal model for medical text and images [5]. These works focus on the accuracy of answers. How well an answer is supported by the image is not reported.

B. Grounding and Visual Prompting

Segmentation and detection models let us link language to pixels. MedSAM adapts Segment Anything to medical images

and generates a mask from a box prompt [6], while YOLO [9] and DETR [8] can supply the boxes. Set-of-Mark prompting showed that numbered marks superimposed on segmented regions allow a model to refer to a region by its number [7]. Our region IDs build on a similar idea, but we use them as a checkable contract rather than as a prompting aid, and we do not accept findings that break it.

C. Hallucination, Uncertainty and Calibration

Self-consistency, in which several outputs are sampled and the majority answer is kept, was introduced for reasoning tasks [12] and gives us a cheap agreement signal. Kuhn et al. first cluster answers with the same meaning before computing the entropy, so that paraphrased answers do not count as disagreement [13]. Calibration is usually summarised by the expected calibration error [14]. We combine agreement, entropy and a localization score into a single gate. Similar cases can also be retrieved with BiomedCLIP embeddings [15] and a FAISS index [16], but this is kept as an optional extra signal.

III. PROPOSED METHOD

A. Pipeline Overview

Given an image I and a question q, the system produces an answer, a list of findings with region IDs, and a confidence label. Fig. 1 shows the six stages. Only the language model is generative. The other stages are deterministic or use fixed models, which keeps the point of failure easy to find.

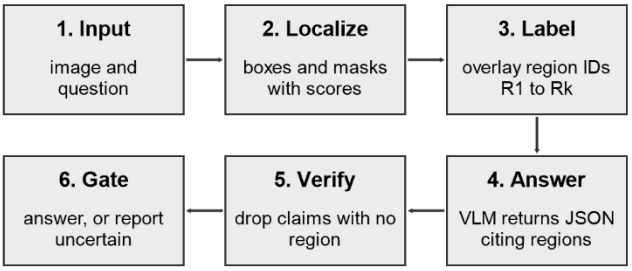


Fig. 1. The six stages of the grounded answering pipeline.

B. Region Proposal

A localizer returns k candidate regions $R = \{r_1, ..., r_k\}$, each consisting of a box b_i, a mask m_i and a score ρ_i. YOLO- or DETR-style detectors propose the boxes and MedSAM turns each box into a mask. To keep the overlay readable, and to stop the model from having to choose between many similar regions, we limit k to a small number. For chest films, the detector can be trained on boxes from RSNA [18] or VinDr-CXR [17]. Other modalities would need a suitable detector, so we treat a second modality as a stretch goal.

C. Region Labelling and Structured Answers

Each region is labelled with its ID (R1 to Rk) and drawn on a copy of the image. This annotated image and the question are given to the VLM, which is instructed to answer only in JSON, following a fixed schema, for example:

```
{"answer": "Yes, right lower zone", "findings":
[{"claim": "effusion", "region_id": "R2", "confidence":
"high"}]}
```

This has two main advantages. The reply can be parsed without guesswork, and each claim can be verified. Both VLMs are given the same prompt text, so any difference in their output is attributable to the models and not to the wording of the prompt.

D. Claim Verification

The parser rejects replies that do not adhere to the schema. A finding is kept only if its region ID names a region in R, and the rest are dropped before anything is shown to the user. During evaluation we go one step further and check the cited region against the ground-truth mask or box. A finding is said to be unsupported when the cited region has an IoU below 0.5 with the annotated structure, which is the typical threshold. The unsupported-claim rate (USR) is computed as

$$USR = N_{\text{unsup}} / N_{\text{claims}} \quad (1)$$

where N_{claims} is the total number of findings emitted by the model. We compute the rate both before and after claim filtering, so that the impact of the filtering is made apparent.

E. Confidence Gate

Three signals are combined. Agreement a is the fraction of N sampled answers that match the majority answer [12]. Normalized entropy H_n is the entropy of the sampled answer distribution divided by log N [13]. The localizer score of the region the answer cites is ρ. The gate calculates

$$c = w_1 a + w_2(1 - H_n) + w_3\rho \quad (2)$$

and gives the answer when $c \geq \tau$, and "uncertain" otherwise. The weights and τ are determined on a held-out split. Raising τ makes the system answer fewer questions with higher accuracy, and we report the full accuracy-versus-coverage curve instead of a single point.

F. Optional Case Retrieval

Case-index images are embedded with BiomedCLIP and searched with FAISS. The nearest cases and their recorded answers can be shown to the user as context, or counted as an extra agreement signal. This increases the workload for data handling and tuning, and is therefore outside the core functionality.

IV. SYSTEM IMPLEMENTATION

A. Architecture

Fig. 2 shows the structure of the application, which consists of four components. The frontend is built with React, Vite and Tailwind CSS. A user can upload an image and type a question. The app uses a canvas to draw region boxes and IDs over the uploaded image. The answer is displayed along with a confidence badge, and if the gate abstains, an "uncertain"

badge is shown instead. A side-by-side view shows the outputs of both VLMs. The backend is a FastAPI application with a single endpoint, POST /analyze. Requests and responses are validated and shaped using Pydantic models, and the application is served with Uvicorn. The model service holds the VLMs, the localizer, the grounding code and the gate. Storage keeps the uploaded images, the saved outputs and the case index.

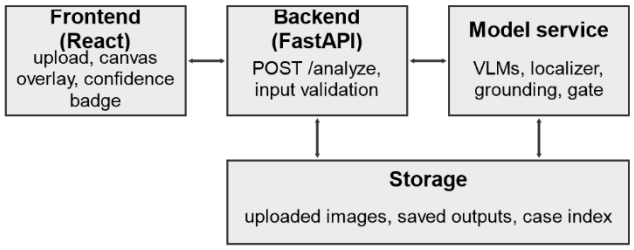


Fig. 2. System architecture of the web application.

B. Models and Deployment

Because these models can use a lot of memory, both Qwen2.5-VL-3B and MedGemma-4B are loaded through Hugging Face Transformers in 4-bit precision using bitsandbytes. They are wrapped in the same code, which takes an image, a prompt and sampling settings, so the rest of the pipeline never has to know which model it is talking to. At run time the request flow is simple. The browser posts the image and question to /analyze, the pipeline runs, and the response includes the answer, the regions and the confidence, which the canvas then draws. The frontend, API and model service are packaged with Docker Compose so that the entire application starts with one command.

V. EVALUATION DESIGN

A. Datasets

Table 1 lists the data. SLAKE is the primary benchmark because it has masks and boxes, and we use its English questions. VQA-RAD is our baseline with a different question style. It has no region annotations, so we use it only for answer metrics. RSNA and VinDr-CXR are used to train and check our localizer on chest X-rays. All evaluation uses test splits, and no private patient data is involved.

Table 1. Datasets and their role.

Dataset	Role	Annotations used
SLAKE [2]	Main benchmark	Questions, masks, boxes
VQA-RAD [1]	Baseline	Questions and answers; no regions
RSNA [18]	Localizer	Boxes for lung opacity
VinDr-CXR [17]	Localizer	Boxes for local findings

B. Models and Conditions

The two VLMs are Qwen2.5-VL-3B and MedGemma-4B. For each, we consider three conditions: free-form answering with no regions, grounded answering under the JSON contract, and grounded answering with the confidence gate. The goal is to separate the effect of grounding from the effect of abstention.

C. Metrics

Table 2 summarises the metrics. Pointing accuracy is the proportion of findings whose cited region reaches an IoU of at least 0.5 with the ground truth. Confidence is assessed by the AUROC of the confidence score for predicting correctness, and by the expected calibration error (ECE), the weighted average gap between accuracy and confidence across confidence bins,

$$ECE = \Sigma_{m=1}^{M} (|B_m|/n) \, |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)| \qquad (3)$$

where predictions fall into M confidence bins B_m, n is the number of predictions, and acc and conf are the accuracy and mean confidence in a bin [14].

Table 2. Planned metrics.

Aspect	Metric
Answers	Accuracy and F1 on closed questions
Grounding	Unsupported-claim rate; pointing accuracy (IoU ≥ 0.5)
Confidence	AUROC; expected calibration error; accuracy versus coverage
Robustness	Change in accuracy and unsupported-claim rate under perturbation
Expert review	Rubric score; Cohen's κ

D. Robustness Tests

We perturb the input with Gaussian noise, contrast changes, JPEG compression, rotations and flips using Albumentations and OpenCV, and we paraphrase some of the questions. A horizontal flip swaps left and right, so if a question asks about a side, such as "what is on the left," the expected answer must be flipped as well. One further test masks out the cited region and asks again. If the answer stays the same, it can be inferred that the cited region was probably not what the model relied on to reach its conclusion.

E. Expert Review and Statistics

Where clinicians are available, they score blinded outputs against a rubric that covers correctness, whether the cited region supports the claim, and whether abstention was appropriate. Cohen's κ is reported for inter-rater agreement [19]. If expert time is not available, an LLM judge will be used as a replacement, and we will disclose this wherever its scores are used. We use bootstrap resampling to generate confidence intervals [20], and paired tests from SciPy for model comparisons.

VI. DISCUSSION AND LIMITATIONS

A. Expectations

We state our expectations as hypotheses, not findings. Grounding is expected to lower the unsupported-claim rate,

probably at the cost of some recall, since true findings that the localizer misses will be dropped. The gate is expected to raise accuracy on the questions it chooses to answer. We do not assume the medical model wins everywhere. If MedGemma-4B beats the general 3B model on accuracy but not on grounding or calibration, or the reverse, that is still a useful result for anyone choosing a model.

B. Limitations

Grounding is not the same as correctness. A model can cite the right region and still misread it, and a lesion the localizer misses makes a true claim impossible to ground. Negative answers, such as "no effusion," are a particular problem for the approach. They claim an absence, so there is no region to point to, and the schema needs a rule for them; we have not settled on one. The approach also inherits the quality of the localizer, and mask quality will vary by modality. Sampling several answers for the gate multiplies inference time, which is of particular concern for an interactive application. The datasets are public research datasets, and the system is not intended for clinical or real patient use. Three-dimensional volumes, video and questions in languages other than English are out of scope.

C. Current Status

This paper covers the design and the evaluation plan. Because the experiments are incomplete, we report no accuracy or calibration figures here; we will include them in the final version.

VII. CONCLUSION

A fluent medical answer is useful only if a reader can check it. We have described a pipeline, together with a React and FastAPI application, that localizes first, makes every claim cite a region, drops claims that cannot be tied to a region, and abstains when it is not confident, and that shows the evidence on the image itself. The evaluation plan considers both a general and a medical-domain VLM, and measures grounding and calibration in addition to accuracy. We expect the results to show where our approach helps and where it does not, and we will discuss both.

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